RE008

RECOMMENDATION PIPELINE ARCHITECTURE REVIEW

Description:
This research will compare the documented recommendation approaches and recommend the most suitable pipeline for Food Remedy's current data, technology and team capabilities.

Tasks:

- Review the documented rule-based, personalised, adaptive and feedback-driven approaches.
- Compare the suitability of rule-based, machine-learning and hybrid pipelines.
- Recommend the most appropriate approach for the current project stage.
- Identify the relevant frameworks, libraries, data structures and prerequisites.
- Provide a high-level implementation and testing guide.
- Produce a concise report with a beginning Summary section.

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Executive Summary

The document proposes a safety-first, hybrid-ready recommendation architecture for the Food Remedy API. It compares rule-based, machine-learning and hybrid pipelines and recommends implementing a deterministic recommendation engine at the current project stage. Strict rules would first exclude foods that conflict with allergies, dietary requirements or validated medical restrictions. Eligible foods would then be ranked using transparent, fixed multi-factor scores based on nutrition, goals, explicit preferences, context, convenience, affordability and variety.

The proposed system contains two user pathways. The main recommendation pathway collects user, food and contextual data; applies the safety rules; calculates nutritional progress; scores and ranks eligible foods; generates structured explanations; and presents the recommendations. The unsafe-food warning pathway is activated when a user independently scans, searches for or selects an unsuitable item, allowing the system to display a warning and suggest a safer alternative.

Stage 1 can be implemented without machine learning using FastAPI, Pydantic, PostgreSQL, SQLAlchemy and Alembic, supported by testing tools such as pytest and Hypothesis. Its core data structures would represent user profiles, food items, health constraints, goals, contextual information, rule outcomes, candidate scores, recommendation results and interaction events.

Machine learning is proposed as a second-stage enhancement after sufficient, reliable interaction data has been collected. It could later support preference learning,

acceptance prediction, collaborative filtering and personalised ranking weights. However, these features would operate only within the soft ranking layer and could never override the deterministic safety rules. Overall, the document recommends delivering a complete, explainable and testable deterministic engine first while recording user interactions to prepare the architecture for future adaptive personalisation.

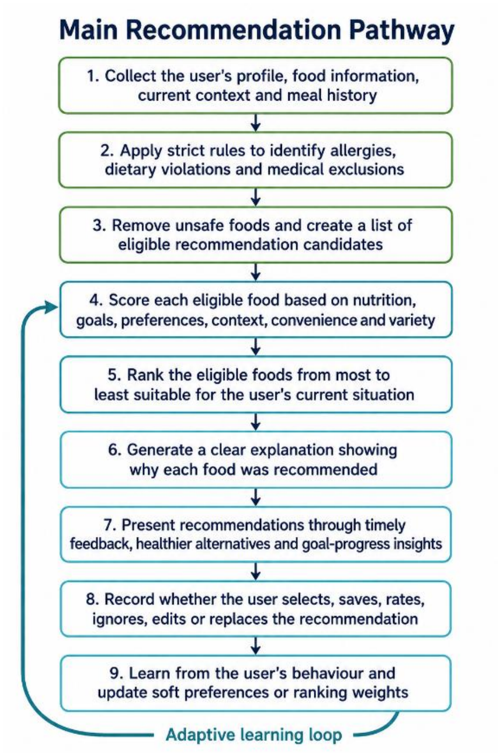
Review of the documented approaches

I have reviewed the documented approached in RE024, RE025, RE026, RE027, RE028 from the T1 research. (add titles to the references and provide them in list form)
This is what I have found.
These documents all together propose a safety-first hybrid recommendation engine that applies deterministic restrictions, ranks eligible foods using personal, nutritional and contextual factors, learns soft preferences from user behaviour, provides just-in-time feedback, and generates traceable explanations for every recommendation.

There are two pathways.

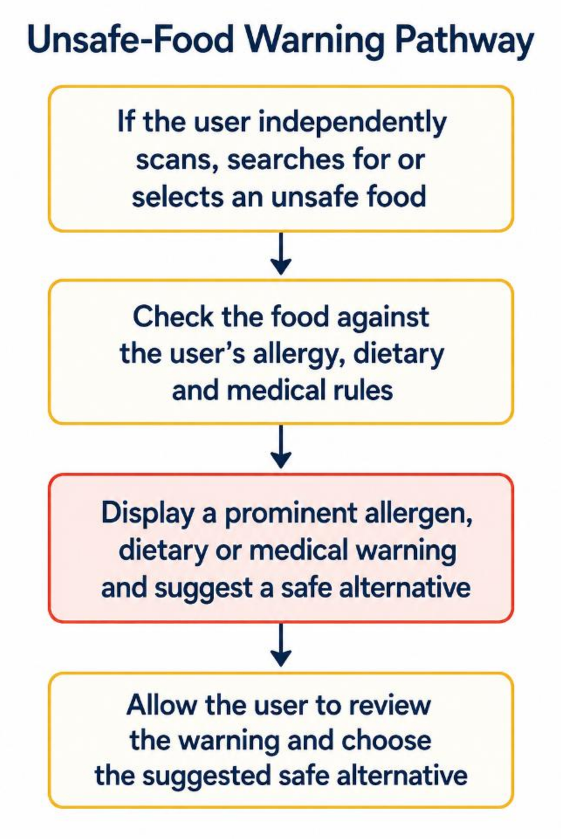
1. **Main Recommendation Pathway** — nine-step recommendation flow with the adaptive-learning loop.

The main recommendation pathway collects the user's profile, food data, current context and meal history before applying strict allergy, dietary and medical rules. Safe foods are then scored and ranked according to factors such as nutrition, goals, preferences and context. The system explains and presents its recommendations, records the user's response, and uses this feedback to improve future rankings without changing the strict safety rules.



2. **Unsafe-Food Warning Pathway** — the separate 4-step warning flow.

The unsafe-food warning pathway is activated when a user independently scans, searches for or selects a food that may be unsuitable. The system checks the food against the user's allergy, dietary and medical rules, displays a clear warning when a conflict is detected, and recommends a safe alternative for the user to review and select.



Comparing suitability of different pipelines

For the current stage of the Food Remedy project, a rule-based pipeline combined with deterministic multi-factor scoring is the most appropriate approach. Strict rules should be used to enforce allergies, dietary requirements and validated medical restrictions, while formulas, thresholds and fixed weights can evaluate nutritional goals, context and user preferences. This approach does not require training data, is easier to test and produces transparent recommendations that can be clearly explained to users.

A purely machine-learning pipeline is not currently suitable because sufficient user-interaction data may not yet be available, and model-based decisions would introduce additional complexity, uncertainty and explainability requirements. However, the system should be designed as a hybrid-ready architecture. User selections, saved foods, ignored suggestions, ratings and meal edits should be recorded from the beginning so that machine-learning features can later be added to the soft ranking layer. These future features may include preference learning, acceptance prediction, collaborative filtering and automatic weight adjustment. Machine learning should improve personalisation but must never override the engine's strict safety rules.

Pipeline approaches comparison

Approach	Suitability for Food Remedy	Advantages	Limitations
Rule-based pipeline	Highly suitable for allergies, diets, medical exclusions, nutrient thresholds and straightforward context rules	Predictable, explainable, testable, safe and requires no training data	Rules must be maintained manually and cannot learn complex preferences
Machine-learning pipeline	Suitable later for preference learning, acceptance prediction and personalised ranking	Learns patterns and can improve personalisation as more data becomes available	Requires sufficient quality data, monitoring and greater explanation effort; must not control safety decisions
Hybrid pipeline	Best long-term approach because it combines strict safety rules with personalised ranking	Preserves safety while allowing recommendations to become more adaptive	More complex to build and maintain; ML components still

Features requiring AI

Component	AI needed now?	Recommended current implementation
Allergy and ingredient checking	No	Deterministic matching rules
Strict diet enforcement	No	Diet and ingredient rules
Medical contraindications	No	Clinically validated rules
Goal and nutrient-deficit calculations	No	Formulas and thresholds
Context matching	No	Fixed rules and scoring adjustments
Multi-factor ranking	No	Transparent weighted scoring

Basic preference handling	No	User-selected preferences and simple interaction counts
Learning complex preferences	Later	Statistical or ML model
Predicting likely acceptance	Later	ML prediction model
Collaborative filtering	Later	ML using behaviour from multiple users
Automatically learning ranking weights	Later	Data-driven optimisation or ML
Explanation generation	No	Structured explanation templates

High – Level Implementation Guide

The implementation of the recommendation system can be split into 2 stages – first priority and second priority.

Stage 1 — Deterministic recommendation engine

Purpose

Build a complete recommendation engine without ML. It will:

1. enforce allergies, diets and medical restrictions;
2. calculate nutritional progress and deficits;
3. apply context-based adjustments;
4. rank safe foods using fixed weights;
5. generate template-based explanations;
6. record interactions for possible future ML.

This is more accurately described as a **rule-based and deterministic scoring pipeline**, rather than purely rule-based.

Stage 2 — Adaptive ML personalisation

Purpose

Stage 2 adds adaptive learning to the soft ranking layer. It may support:

- preference learning;

- acceptance prediction;
- collaborative filtering;
- personalised ranking weights;
- behavioural pattern detection;
- progress-sensitive recommendations.

The Stage 1 safety gate remains mandatory. ML receives only eligible foods and cannot override allergies, diets or medical exclusions.

However, for now let's focus only on stage 1.

Recommended frameworks and libraries

Area	Recommended technology	Purpose
API framework	FastAPI	Recommendation endpoints and automatic API documentation
Input validation	Pydantic	Validate profiles, foods, context and response structures
Database	PostgreSQL	Store users, foods, rules, goals and interactions
Database access	SQLAlchemy	Map application objects to relational tables
Schema migrations	Alembic	Version and safely update the database schema
Unit/integration testing	pytest	Test rules, calculations, ranking and endpoints
Property-based testing	Hypothesis	Generate edge cases and verify safety invariants
API testing	HTTPX/FastAPI TestClient	Test complete API requests and responses
Configuration	Environment variables and versioned JSON/YAML	Store weights, thresholds and rule settings
Optional caching	Redis	Cache food candidates or frequently requested results

Required data structures

Structure	Important fields	Purpose
UserProfile	allergies, diets, medical restrictions, explicit preferences	Permanent or relatively stable user information
FoodItem	ingredients, allergens, diet tags, nutrients, price, preparation time	Information used to filter and rank foods
HealthConstraint	nutrient, operator, threshold, severity, source, version	Validated medical or nutritional rule
GoalTarget	calories, protein, carbohydrates, fat, sugar, sodium	Daily or meal-level targets
ContextSnapshot	time, meal type, activity, consumed nutrients, availability	Current recommendation situation
RuleResult	passed, rule ID, reason code, severity	Result of evaluating one strict rule
CandidateScore	factor scores, weights, total score, explanation codes	Transparent scoring result for one food
RecommendationResult	ranked foods, explanations, rule version, weight version	Complete API response
InteractionEvent	impression, selection, save, ignore, replacement, timestamp	Future learning data

Stage 1 – High level pipeline implementation